

HR ANALYTICS PORTFOLIO

HR attrition is measurable, explainable, and actionable

One end-to-end analytics project turns workforce data into practical retention priorities.

THE PROJECT

One project connects data cleaning to business action

01

1,470 raw records

Across 37 columns.

03

Power BI

Make patterns visible in a dashboard.

Focus: workforce composition, attrition, salary, satisfaction, roles, and workload.

02

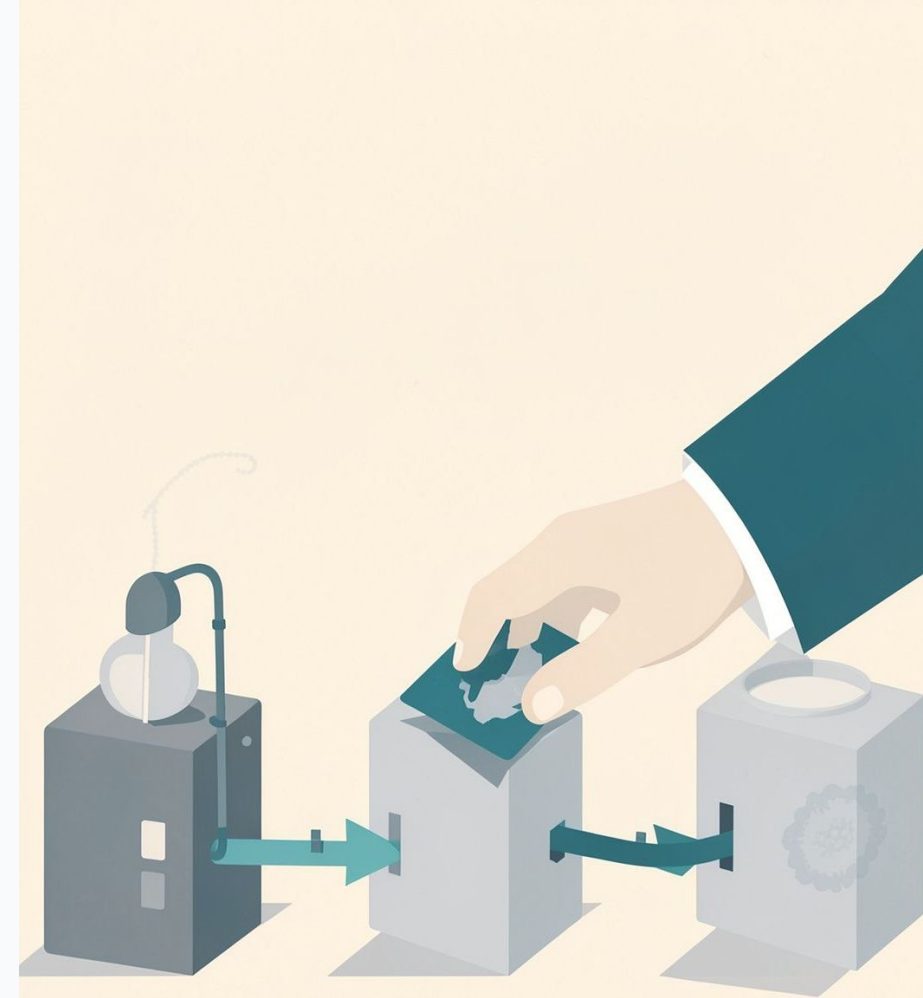
Python → PostgreSQL

Clean the data, then structure the analysis.

04

Recommendations

Translate evidence into retention priorities.



Cleaning removed six duplicate employee records before reporting

01

EmpID

EmpID was treated as the employee-level identifier.

02

–6 duplicates

Duplicate employee records were removed before reporting.

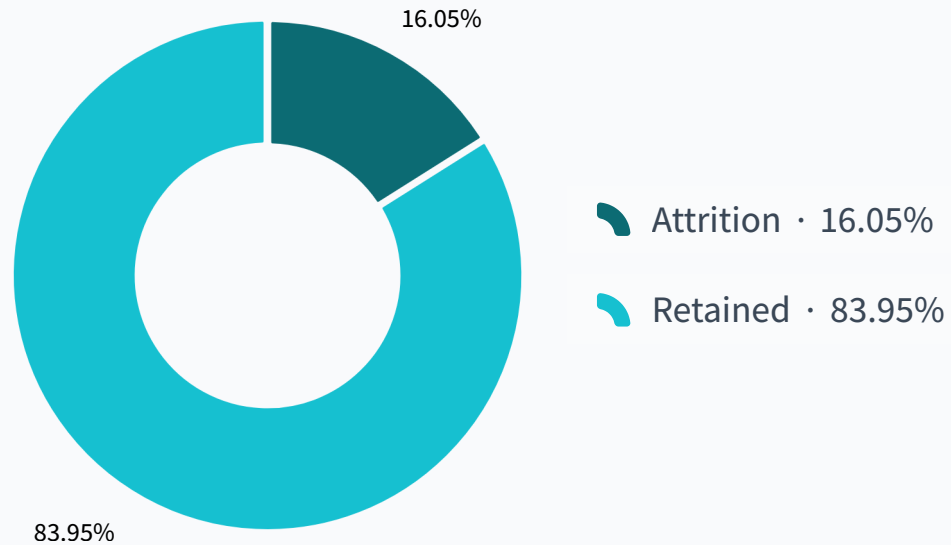
03

1,464 unique employees

Missing values and data types were reviewed before export to hr_01.csv.

WORKFORCE BASELINE

The workforce baseline is 1,464 employees and 16.05% attrition



Employees

1,464

Attrition

16.05%

Cases

235

Average age

36.98

Average experience

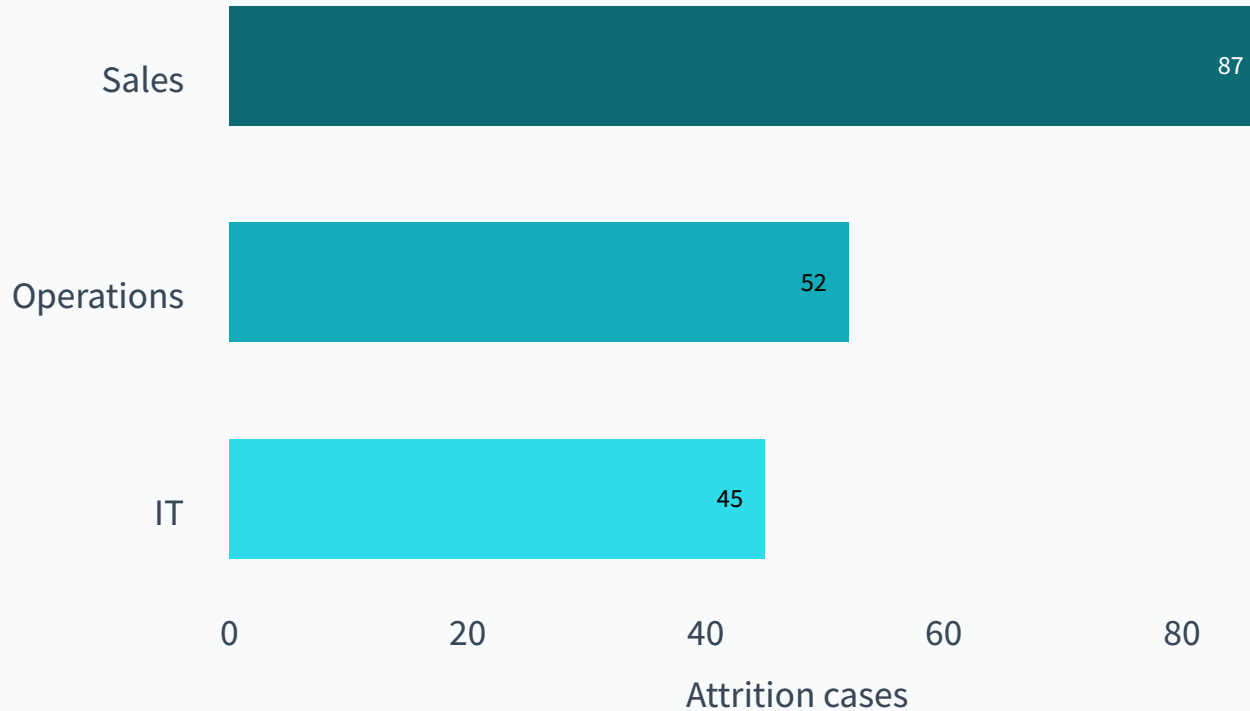
7.02

The baseline establishes the scale of the retention problem before prioritization begins.

DEPARTMENT SIGNAL

Sales is the first retention priority

Department



Share of reported attrition

37.02
%

87 Sales attrition cases

Investigate targets, incentives, manager support, and work culture.

ROLE-LEVEL RISK

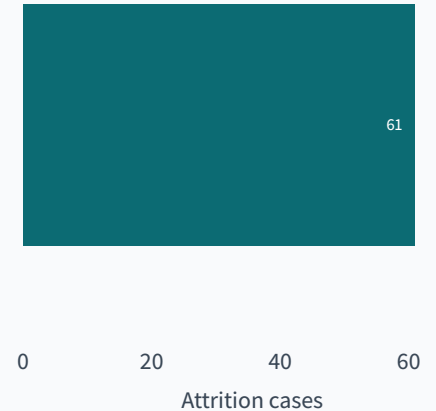
Laboratory Technicians carry the highest role-level risk

Lower satisfaction levels account for notable portions of the cases.

Review pay, workload, training, and progression for technical entry-level roles.

Role

Laboratory Technician



Workload and compensation are connected retention levers

WORKLOAD

Overtime is a visible pressure signal

Employees working overtime represent a substantial share of attrition cases.

Intervention: staffing and workload redistribution.

COMPENSATION

Lower and mid salary bands are important segments

Compensation belongs in the retention diagnosis, alongside workload.

Intervention: salary benchmarking and promotion timelines.

The largest workforce segment needs a visible growth path

LARGEST WORKFORCE GROUP

26-35

The segment to keep growing with the organization.



Mentoring



Skill
development



Internal
mobility



Recognition

Cross-validation makes the findings decision-ready

01

Python

Validates cleaning and patterns.

02

SQL

Structures the analysis.

03

Power BI

Makes trends visible.

The strongest visuals are department attrition, role risk, overtime, and salary bands.

RECONCILED HEADCOUNT

1,464

Headcount reconciles after duplicate removal.

Turn the analysis into a focused retention agenda

- 01 Investigate Sales pressure and incentives.
- 02 Stabilize Laboratory Technician pay, workload, and progression.
- 03 Reduce excessive overtime through workforce planning.
- 04 Track satisfaction and retention by role, department, salary, and age group.